Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati

A Mid term report for the BDM capstone Project

Submitted by

Name: Rudra Rajeswari

Roll number: 24f1001011

Email: 24f1001011@ds.study.iitm.ac.in

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1. Executive Summary and Title ……4

2. Proof of Originality of the Data …….4 - 5

3. Metadata and Descriptive Statistics ……...5 - 8

4. Detailed Explanation of Analysis process ……..8 - 10

5. Results and Findings ……..10 - 12

Declaration Statement

I am working on a Project Titled "Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati **".** I extend my appreciation to Meridian, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

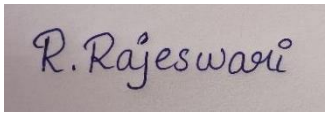
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of the candidate:



Name: Rudra Rajeswari

Date: 08-01-2026

Data-Driven Optimization for Operational Efficiency at Meridian Restaurant, Tirupati

1. Executive Summary

Meridian Restaurant is a small owner operated restaurant, which was opened in 2020 by Mr. Gopi Krishna in Vidyanagar, Kota, Tirupati District, Andhra Pradesh. The restaurant is located between 11:30 AM and 9:30 PM and provides students, families, and local residents with low prices on vegetarian and non-vegetarian food. The company employs 10-12 people to work on rotational shifts and has been witnessing a steady growth but has been fully depending on manual record keeping in form of handwritten notebooks. The project solutions will solve three major issues in business, including workforce optimization and resource planning, inventory and ingredient management, and general business performance analysis.

The data used in this research is five months of operational data (August 1st -December 31st 2025), which comprises of the most important metadata of the day, such as daily sales revenue, number of customers, veg orders, non-veg orders, total orders, amount of orders in afternoon, and night, and customer preferences. The data were tabulated with the help of MS Excel, sorted out to eliminate irregularities and summarized with the help of descriptive statistics in terms of mean, totals, percentages and comparison of trends. Some basic survey inputs and observations were also taken to get insights into customer preferences and hours of peak.

The methodology involves descriptive analytics to summarize sales and workforce trends, diagnostic analytics to understand the causes of inefficiencies and predictive analytics with trend-based forecasting to provide an approximate estimate of inventory needs and staffing needs in future. Basic visualization (bar charts, line graphs, and pie charts), and MS Excel tools were utilized. Intermediate findings show that there is a definite peak business time and high demand menu items, frequency been wasted, and mismatch in staff during the weekends, which can be used to understand what can be done to improve the operational planning.

2. Proof of Originality

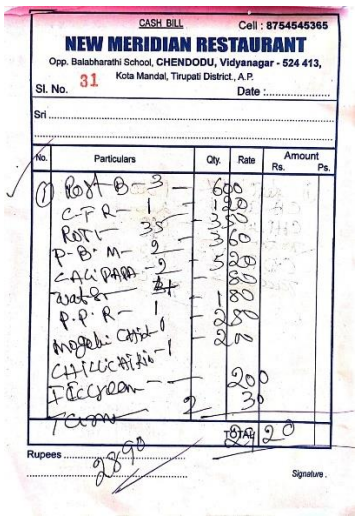
- Video interaction clip with the owner: https://drive.google.com/file/d/15Cb51ngMJuVtAxikLqgMZRZjFpkKdehQ/view?usp=sharing
- Letter from organization:

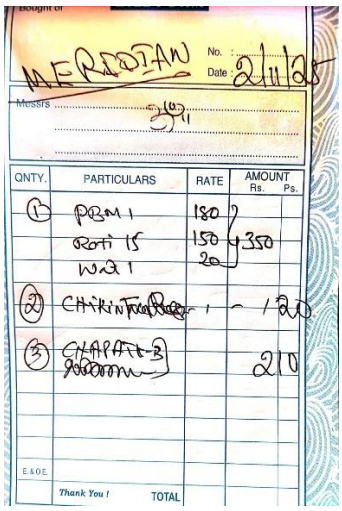
https://docs.google.com/document/d/1cS5UvbzSdOZC1qafa7iXtWZat8hbPcwA/edit?usp=sharing&ouid=117388886427818226244&rtpof=true&sd=true

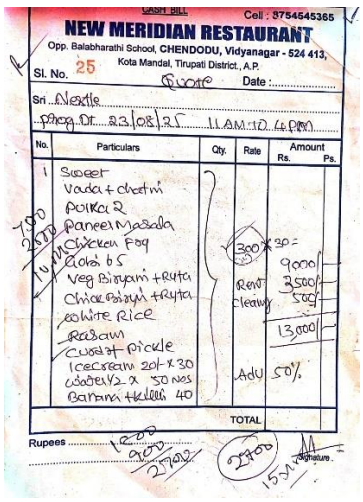
Note: As this is a small- scale Restaurant , they do not have stamp

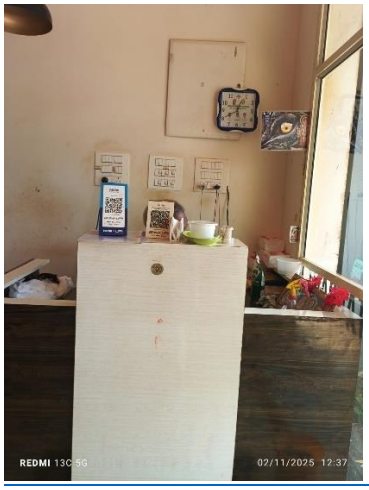
- Dataset link: https://docs.google.com/spreadsheets/d/1wo302wrbLe1HludZXvqLqqM2RNkmDSdL/edit?usp=sharing&ouid=117388886427818226244&rtpof=true&sd=true

- Images of manual registers and organization view:









3. Metadata and Descriptive statistics

3.1 Metadata

The data for this project was collected from Meridian Restaurant in Tirupati. It includes records from August 2025 to December 2025. The restaurant maintains its daily sales and order details in a book register. This data was manually entered into MS Excel for easy analysis. Excel was used to study sales, customer

preferences, and peak order times. This data helps in understanding staff needs, inventory usage, and overall business performance.

- **Date** : Represents the calendar date from 01-08-2025 to 31-12-2025. This variable helps in performing time-series analysis and understanding monthly and seasonal business trends.
- **Day** : Indicates the day of the week (Monday to Sunday). It is useful for identifying differences in customer flow and food preferences between weekdays and weekends.
- **Families Served** : Refers to the number of families served per day. This variable shows customer footfall and helps analyze restaurant capacity utilization and staffing needs.
- **Veg Orders** : Represents the total number of vegetarian food orders per day. It helps in understanding vegetarian customer preferences and planning the purchase of veg ingredients.
- **Non-Veg Orders** : Represents the total number of non-vegetarian food orders per day. This variable is important for predicting demand for non-vegetarian ingredients and controlling inventory.
- **Total Orders** : It is the sum of veg and non-veg orders placed per day. This indicates the overall business volume and daily workload on staff.
- **Daily Revenue (₹)** : Represents the total revenue generated per day in rupees. It is a key measure of financial performance and business growth.
- **Peak Order Time** :Indicates the time of day when the maximum orders are placed, categorized as Afternoon or Night. This helps identify peak operational hours for efficient workforce scheduling and service planning.

C21 | 11

Date	Day	Families Served	Veg Orders	Non-Veg Orders	Total Orders	Daily Revenue (₹)	Peak Order Time
01-09-2025	Monday	9	20	20	40	10518	Night
02-09-2025	Tuesday	9	15	21	36	8081	Afternoon
03-09-2025	Wednesday	9	23	16	39	8543	Afternoon
04-09-2025	Thursday	8	22	20	42	9045	Afternoon
05-09-2025	Friday	8	17	25	42	9285	Night
06-09-2025	Saturday	15	21	18	39	11356	Night
07-09-2025	Sunday	13	20	16	36	9160	Afternoon
08-09-2025	Monday	9	17	17	34	7753	Afternoon
09-09-2025	Tuesday	9	14	21	35	7530	Night
10-09-2025	Wednesday	10	23	29	52	10533	Afternoon
11-09-2025	Thursday	10	15	30	45	10331	Night
12-09-2025	Friday	10	20	21	41	10363	Afternoon
13-09-2025	Saturday	15	15	23	38	10994	Afternoon
14-09-2025	Sunday	11	22	30	52	14659	Night
15-09-2025	Monday	8	14	17	31	7338	Afternoon
16-09-2025	Tuesday	8	17	25	42	9775	Afternoon
17-09-2025	Wednesday	8	24	30	54	11752	Afternoon
18-09-2025	Thursday	8	21	28	49	11439	Afternoon
19-09-2025	Friday	9	18	16	34	7035	Night
20-09-2025	Saturday	11	23	21	44	10659	Afternoon
21-09-2025	Sunday	13	20	16	36	7957	Afternoon
22-09-2025	Monday	9	16	23	39	10063	Afternoon
23-09-2025	Tuesday	8	22	28	50	10440	Night
24-09-2025	Wednesday	8	21	16	37	8667	Afternoon
25-09-2025	Thursday	10	21	25	46	9775	Night
26-09-2025	Friday	9	17	18	35	7642	Night
27-09-2025	Saturday	13	21	22	43	11375	Afternoon
28-09-2025	Sunday	12	19	24	43	10732	Afternoon
29-09-2025	Monday	10	16	19	35	7080	Night
30-09-2025	Tuesday	10	18	18	36	8391	Afternoon

September 2025 | October 2025 | November 2025 | December 2025 | analysis | Analysis Summary

Fig1: Monthly data

3.2 Descriptive Statistics

Descriptive statistics were used to summarize the data collected from Meridian Restaurant, Tirupati, for the period 1st August 2025 to 31st December 2025. The analysis focuses on key variables such as families served, veg orders, non-veg orders, total orders, and daily revenue to support workforce planning, inventory management, and business performance analysis.

Table 1: Measures of Central Tendency

Variable	Mean	Median	Mode
Families Served	10.39	10	9
Veg Orders	18.65	18	14
Non-Veg Orders	24.03	24	30
Total Orders	42.68	44	45
Daily Revenue (₹)	8,968.74	8,931	5,869
Peak Order Time	Night	Night	Night

- The mean and median values are closely aligned, indicating stable daily business operations.
- Non-veg orders have a higher mean and mode, showing consistent customer preference.
- The average daily revenue (~₹9,000) aligns with the restaurant's reported income range.

Table 2: Measures of Variability

Variable	**Standard Deviation**	**Minimum**	**Maximum**
Families Served	1.89	8	14
Veg Orders	3.32	14	24
Non-Veg Orders	4.72	16	30
Total Orders	6.17	33	53
Daily Revenue (₹)	1,396.80	5,869	10,988

- Low standard deviation in families served indicates predictable daily footfall.
- Higher variability in non-veg orders suggests demand spikes on specific days.
- Revenue variability reflects weekend and peak-hour sales growth.

The measures of central tendency show stable daily customer footfall, order volume, and revenue, indicating predictable business operations. This helps in optimizing workforce and shift planning by allocating staff according to average demand. The higher average and variability in non-vegetarian orders highlight the need for efficient inventory planning and ingredient control to avoid shortages or wastage. The variability in total orders and revenue reflects peak and off-peak periods, enabling management to focus resources on high-demand hours and best-selling items to improve overall business performance.

4. Detailed Explanation of Analysis Process

4.1. Data Cleaning and Preprocessing

Due to the fact that the data used in this study had to be manually transferred by a physical restaurant register to Microsoft Excel, there was a need to undertake a comprehensive data cleaning and preprocessing process to guarantee the accuracy, consistency and reliability of the analysis.

- The initial step was the elimination of duplicates whereby only one valid entry was made per date and this served to avoid overestimation of orders and revenue.
- Then missing values in order counts or revenue were treated by cross-checking with original register; values could not be available; then reasonable averages of similar dates were taken not to disrupt any trend.
- In order to ensure consistency, standardization of data types was done by standardizing all dates in the DD-MM-YYYY format and making all numeric fields including orders and revenue as numerical values.
- Then validation checks were implemented to make sure that there was internal consistency, especially that the overall number of orders was the same as the sum of vegetarian and non-vegetarian orders.
- Value of revenue was also verified to ensure that it was within realistic daily operation limits of Rs.7,000 to Rs.10,000, which also helped in detection and correction of mistakes in data entry.
- Also, order times were divided into standardized time slots (Afternoon and Night) to facilitate the analysis of peak hours and facilitate workforce planning.

4.2. Analysis Process / Method

1. Revenue Trend Analysis (Monthly)

- Line chart using average daily revenue for each month (Aug–Dec 2025).

- Average Daily Revenue $= \frac{\sum \text{Daily Revenue}}{\text{Number of Days}}$
- To identify growth, decline, and seasonal variation in revenue.
- Line charts are suitable for time-series data and clearly show trends compared to tables or raw numbers.
- Supports Business Performance Analytics by identifying high- and low-performing months for better planning.

2. Veg vs Non-Veg Order Pattern Analysis

- Bar chart comparing *average veg orders* and *average non-veg orders* month-wise.
- Average Orders $= \frac{\sum \text{Orders}}{\text{Number of Days}}$
- To understand customer food preferences.
- Bar charts allow easy comparison between categories, which is more effective than trend or regression analysis for preference study.
- Supports Inventory & Ingredient Control by aligning ingredient procurement with demand patterns.

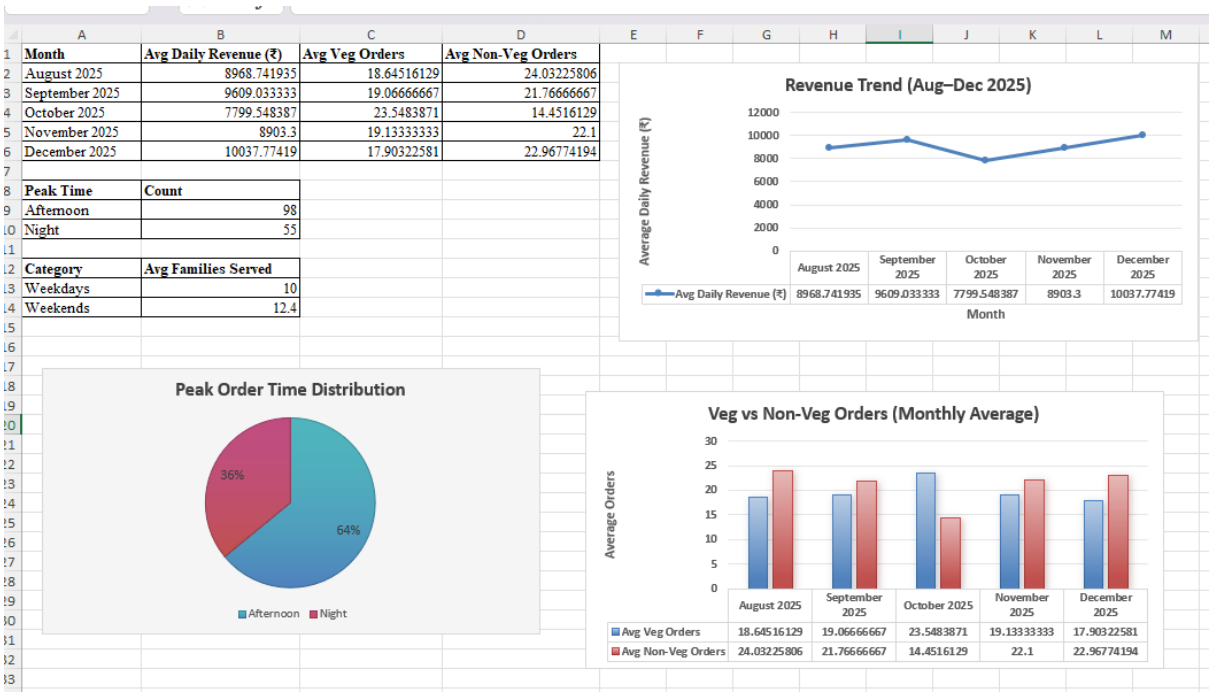
3. Peak Order Time Analysis

- Pie chart showing distribution of orders between Afternoon and Night.
- Percentage $= \frac{\text{Orders in Time Slot}}{\text{Total Orders}} \times 100$
- To identify peak business hours.
- Pie charts are effective for proportional comparison and are simpler than time-series analysis for small datasets.
- Supports Workforce & Resource Planning by helping schedule staff during peak hours.

4. Weekend vs Weekday Footfall Analysis

- Column chart comparing average families served on weekdays and weekends.
- Avg Families Served $= \frac{\sum \text{Families}}{\text{Number of Days}}$
- To assess variation in customer footfall.
- Column charts clearly highlight differences between discrete categories.
- Supports helping allocate additional staff and seating on weekends.

➢ **Summarized Dataset & Analysed Sheet Link**: https://docs.google.com/spreadsheets/d/1wo302wrbLe1HludZXvqLqqM2RNkmDSdL/edit?usp=drive_link&ouid=117388886427818226244&rtpof=true&sd=true



Month	Avg Daily Revenue (₹)	Avg Veg Orders	Avg Non-Veg Orders
August 2025	8968.741935	18.64516129	24.03225806
September 2025	9609.033333	19.06666667	21.76666667
October 2025	7799.548387	23.5483871	14.4516129
November 2025	8903.3	19.13333333	22.1
December 2025	10037.77419	17.90322581	22.96774194

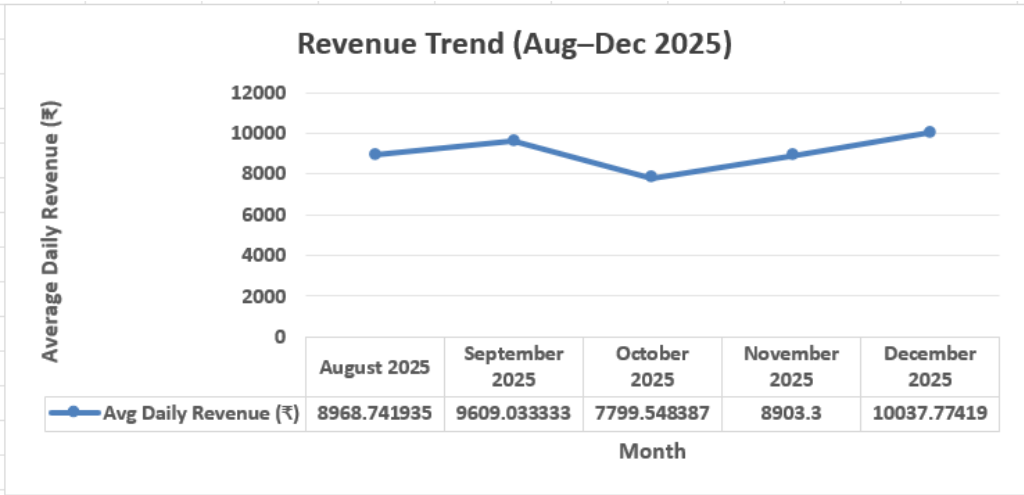
Peak Time	Count
Afternoon	98
Night	55

Category	Avg Families Served
Weekdays	10
Weekends	12.4

Fig2: Analyzed sheet

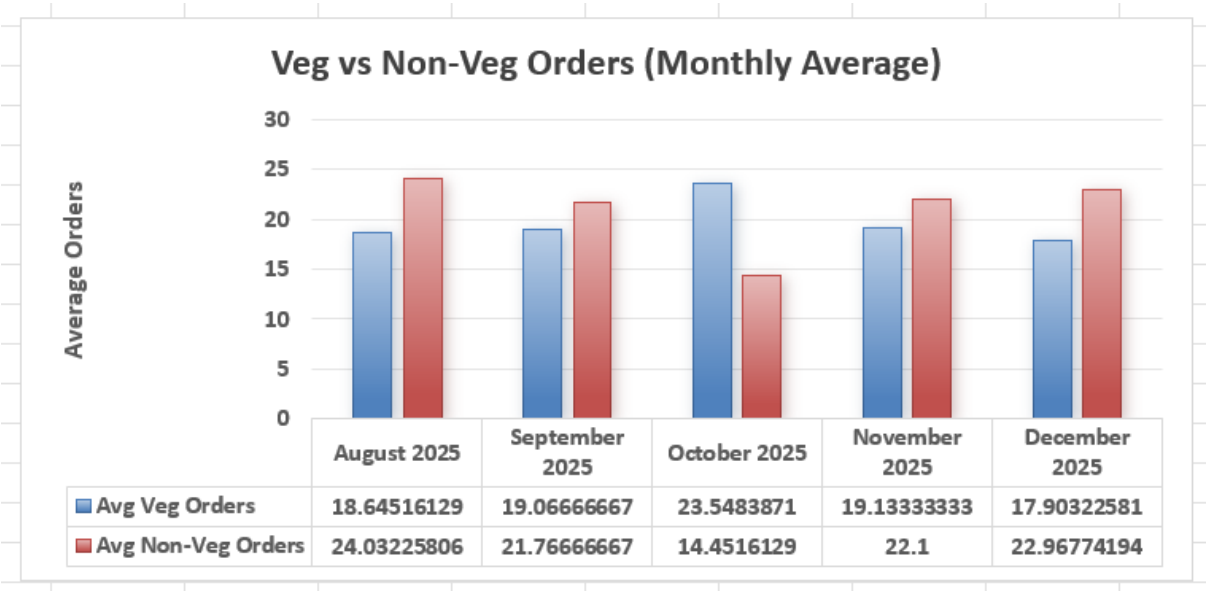
5. Results and Findings

5.1. Revenue Trend Analysis (Aug-Dec 2025)



Trend of the revenues reveals that it experienced a slow rise between August and September and then a decrease in October and again steadily increasing in November and December. The decline in October could be attributed to decrease in the number of customers or seasonal influences. The largest average daily revenue is recorded in December, which means that it is in demand in the festive or end-of-year seasons. The trend demonstrates the existence of seasonal variations in the restaurant revenue. In general, the performance in revenue increases during the end of the year.

5.2. Veg vs Non-Veg Orders Analysis



	August 2025	September 2025	October 2025	November 2025	December 2025
Avg Veg Orders	18.64516129	19.06666667	23.5483871	19.13333333	17.90322581
Avg Non-Veg Orders	24.03225806	21.76666667	14.4516129	22.1	22.96774194

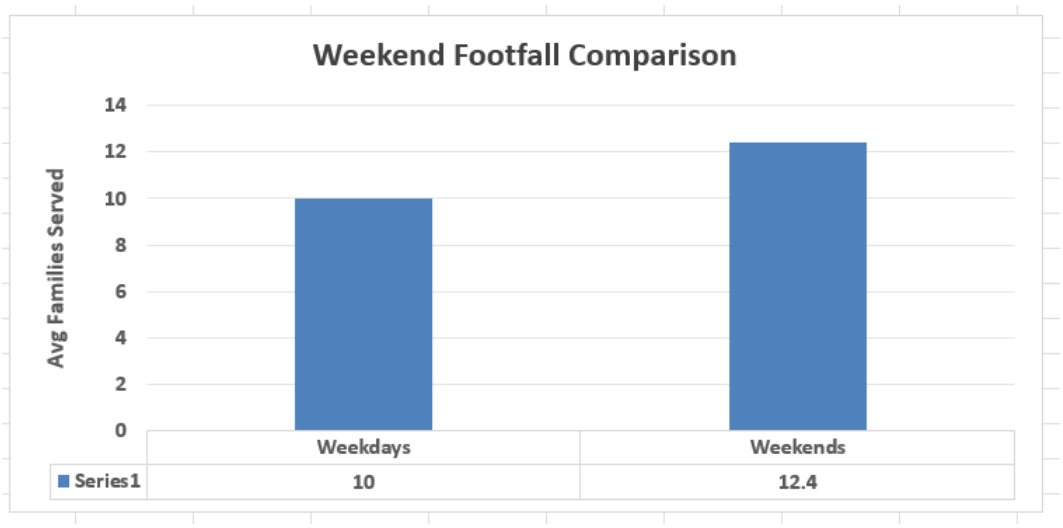
The examination of the non-vegetarian orders versus the vegetarian orders reveals that the non-vegetarian orders are always higher in most months. In October, there is a temporary change of vegetarian orders and non-veg orders whereby it increased and decreased respectively, which indicates a shift in customer preference during this month. During other months, the non-veg demand prevails and this is adding to the total sales. This trend underscores the need to have sufficient non-veg stocks. The findings are an excellent portrayal of customer preferences in food.

5.3. Peak Order Time Distribution

The peak order time analysis depicts that about 64 percent of the orders are done during the afternoon and 36 percent at night. This implies that the afternoons are the busiest hours of the day. The increased afternoon demand can be as a result of the lunch time customers and family visits. This trend indicates that additional

manpower would be needed in the afternoon. Effective schedule arrangement will enhance quality of service during peak periods.

5.4. Weekend vs Weekday Footfall Comparison



The footfall on the weekends is much more than the weekdays whereby on the weekends the average number of families is 12.4 and on weekdays is 10. This means that the number of customers is higher on weekends. Leisure dining and family outings might have contributed to the increase in the number of people visiting. This trend highlights the importance of the increased staffing and resources at weekends. It can assist in controlling the movement of crowds and efficiency in services through proper planning.

The overall analysis shows that the demand of the customers fluctuates depending on time, type of order, and days of the week. Weekends and peak periods also pose a challenge as they raise the revenues and the number of people visiting, thus necessitating the planning of flexible workforce. Non veg items contribute significantly towards sales and inventory. The variability of the revenue in seasons means that there is a possibility of special promotions. These results are a solid foundation of operational enhancement.